

# VANESSA SKLYAROVA

vanessasclearova@gmail.com ◇ Web: <https://vanessik.github.io/> ◇ Scholar: [7ez3M0AAAAAJ](#)

Computer Vision & AI researcher with 6+ years of experience working on diverse research & engineering projects. At the moment, a Ph.D. student at ETH Zurich, ([CVG group](#)) and Max Planck Institute for Intelligent Systems ([Perceiving Systems](#)) with a focus on digital humans. Five first-author papers at top venues including **ICCV 2023 (Oral, top 1.8%)** and **3DV 2026 (Best Paper Runner-Up)**.

## PROFESSIONAL EXPERIENCE

---

### Meta Zurich, ([Reality Labs](#))

Research Scientist Intern

06/2026 - Present

- Research on video diffusion models.

### ETH Zurich, ([CVG group](#))

Ph.D. Researcher

07/2025 - Present

- Research on dynamic head avatars from multi-view images.

### Max Planck Institute for Intelligent Systems, ([Perceiving Systems](#))

Ph.D. Researcher

06/2023 - Present

- 3D reconstruction, rendering, and simulation of head avatars with strand-based hair.
- Developed a text-guided generative model and single-view strand reconstruction method.
- Research on hair rendering with Gaussian Splatting.
- Published 4 first-author papers (CVPR 2024, ICCV 2025, 3DV 2026, ECCV 2026) + 7 collaborative works.
- Supervised 5 students and 4 group projects on computer vision and virtual reality.

### Samsung AI Center

Research Engineer

08/2021 - 06/2023

- Worked on mesh reconstruction methods for learning human hair geometry;
- Co-authored a paper accepted to ECCV 2022 on one-shot mesh reconstruction of human avatars;
- Co-authored a patent on reconstruction and rendering of human head avatars ([#12169900](#));
- Worked on multi-view hair reconstruction methods with strand-based representations;
- Co-authored a paper accepted to ICCV 2023 on strand-based hair reconstruction for human avatars.

### Samsung AI Center

Assistant Engineer

06/2021 - 08/2021

- Developed a Reptile-based meta-learning framework for NeRF models;
- Integrated the meta-learning pipeline into the top-performing model in distributed mode.

### Huawei

Assistant Engineer

06/2020 - 10/2020

- Tested state-of-the-art low-bit quantization methods on Transformer and CNN/MLP models;
- Conducted experiments on automatic quantization using reinforcement learning.

### Gazprom Neft

Assistant Engineer

02/2019 - 01/2020

- Built RL agents (DQN, DQfD) and contextual bandits for dynamic pricing, achieving 90%+ of optimal rewards;
- Co-authored patent on dynamic pricing ([#2022619566](#));

## EDUCATION

### ETH Zurich

Ph.D. in Computer Science

Zurich, Switzerland

09/2023 - Present

Topic: *Digital human avatars*, prof. Marc Pollefeys, prof. Michael Black, prof. Justus Thies

### Skolkovo Institute of Science and Technology, [Skoltech](#)

Moscow, Russia

M.Sc. in Data Science (GPA 4.84 / 5.00)

Topic: *Reinforcement learning for Deep brain simulation* ([thesis](#)), prof. Dmitry Dyllov

### Moscow Institute of Physics and Technology, [MIPT](#)

Moscow, Russia

M.Sc. in Applied Mathematics and Physics (GPA 4.95 / 5.00)

### Moscow Institute of Physics and Technology, [MIPT](#)

Moscow, Russia

B.Sc. in Applied Mathematics and Physics (GPA 4.4 / 5.00)

Topic: *Mathematical model of HIV infection*, Institute of Numerical Mathematics, prof. Gennady Bocharov

## RELEVANT PUBLICATIONS & PREPRINTS

### Head Avatars with Dynamic Explicit Hair

2026

V. Sklyarova\*, H. Chen\*, B. Kabadayi, T. Kirschstein, Z. Fan, X. Wang, G. Pons-Moll, M. Nießner, M. Pollefeys, M. Black, J. Thies

ECCV

### MOCHI: Registration-free Multi-view Face Prediction ([page](#))

2026

P. Filntisis, G. Retsinas, R. Danecek, [V. Sklyarova](#), P. Maragos, T. Bolkart

CVPR

### PhysHead: Simulation-Ready Gaussian Head Avatars ([page](#))

2026

B. Kabadayi, [V. Sklyarova](#), W. Zielonka, J. Thies, G. Pons-Moll

CVPR

### GeomHair: Reconstruction of Hair Strands from Colorless 3D Scans ([page](#))

2026

R. Lazuardi, A. Sevastopolsky, E. Zakharov, M. Nießner, [V. Sklyarova](#)

CVPRW

### How to Build Digital Humans? From Priors to Photorealistic Avatars ([page](#))

2026

W. Zielonka\*, T. Kirschstein\*, T. Bolkart\*, S. Giebenhain, [V. Sklyarova](#), X. Deng, et al.

Eurographics (STAR)

### Vid2Haircut: 3D Strand-Based Hairstyle Reconstruction from Video ([page](#))

2026

F. Ben Ayed, G. Becherini, J. Thies, [V. Sklyarova](#)

Eurographics (Short Papers), CVPRW

### NeuralFur: Animal Fur Reconstruction from Multi-view Images ([page](#))

2026

[V. Sklyarova\\*](#), B. Kabadayi\*, A. Yiannakidis, G. Becherini, M. Black, J. Thies

3DV

(Oral presentation, Best Paper Runner Up Award)

### Im2Haircut: Single-view Strand-based Hair Reconstruction for Human Avatars ([page](#))

2025

[V. Sklyarova](#), E. Zakharov, M. Prinzler, G. Becherini, M. Black, J. Thies

ICCV

### Joker: Conditional 3D Head Synthesis with Extreme Facial Expressions ([page](#))

2025

M. Prinzler, E. Zakharov, [V. Sklyarova](#), B. Kabadayi, J. Thies

3DV

### Gaussian Haircut: Human Hair Reconstruction with Strand-Aligned 3D Gaussians ([page](#))

2024

E. Zakharov, [V. Sklyarova](#), M. Black, G. Nam, J. Thies, O. Hilliges

ECCV

### HAAR: Text-Conditioned Generative Model of 3D Strand-based Human Hairstyles ([page](#))

2024

[V. Sklyarova](#), E. Zakharov, O. Hilliges, M. Black, J. Thies

CVPR

### Neural Haircut: Prior-Guided Strand-Based Hair Reconstruction ([page](#))

2023

[V. Sklyarova](#), J. Chelishev, A. Dogaru, I. Medvedev, V. Lempitsky, E. Zakharov

ICCV

(Oral presentation, top 1.8% of the submissions)

### Realistic one-shot mesh-based head avatars ([page](#))

2022

T. Khakhulin, [V. Sklyarova](#), V. Lempitsky, E. Zakharov

ECCV

\* denotes equal contribution.

## SELECTED AWARDS

---

### Best paper runner-up award

*NeuralFur: Animal Fur Reconstruction from Multi-view Images*

Vancouver, Canada

March 2026

### Outstanding Reviewer Award

CVPR 2026

### Best presentation runner-up award

*Text-Conditioned Generative Model of 3D Strand-based Human Hairstyles*

Erlangen, Germany

Dec 2023

## SERVICE & COMMUNITY CONTRIBUTIONS

---

**Workshop Organizer.** MUSTCV Workshop, ECCV 2026; CV4Metaverse Workshop, CVPR 2025.

**Conference Reviewer.** CVPR, ICCV, ECCV, SIGGRAPH, SIGGRAPH Asia, 3DV, Eurographics.

**Student Supervision.** Supervised 5 M.Sc. students, 1 B.Sc. student, and 4 group projects for the Computer Vision and Virtual Reality course at ETH Zurich. Co-authored two student-led papers as senior author (Vid2Haircut, GeomHair).

## TEACHING

---

**3D Vision**, ETH Zurich, CVG Lab

Spring 2026

**Linear Algebra**, ETH Zurich

Autumn 2025

**Seminar on Human-Performance Capture**, ETH Zurich, AIT Lab

2024 (2 semesters)

## PATENTS

---

**Reconstruction and Rendering of Human Head Avatars**, U.S. Patent ([#12169900](#)). Co-inventor.

**Dynamic Pricing System**, Russian Patent ([#2022619566](#)). Co-inventor.

## SELECTED PRESENTATIONS

---

### Keynote

*"Modeling Strand-based Hairstyles for Digital Human Avatars"*

P3HA workshop

CVPR 2026

### Oral Presentation

*"NeuralFur: Animal Fur Reconstruction From Multi-View Images"*

Vancouver, Canada

3DV 2026

### Invited Talk

*"Modeling Realistic Hairstyles for Digital Human Avatars"*

ETH AI Center

Jan 2026

### Talk

*"Text-Conditioned Generative Model of 3D Strand-based Human Hairstyles"*

FAU University, Erlangen

Dec 2023

### Talk

*"Generating Realistic 3D Hair Geometry for Human Avatars"*

CVG Lab, ETH Zurich

May 2024

### Invited Talk

*"Generating Realistic 3D Hair Geometry for Human Avatars"*

Google Zurich

Feb 2024

### Oral Presentation

*"Neural Haircut: Prior-Guided Strand-based Hair Reconstruction"*

Paris, France

ICCV 2023

### Talk

*"Neural Haircut: Prior-Guided Strand-based Hair Reconstruction"*

CV4Metaverse Workshop

ICCV 2023

### Talk

*"Neural Haircut: Prior-Guided Strand-based Hair Reconstruction"*

Amazon, Tübingen

July 2023

## LANGUAGES

---

English (full professional), German (basic), Russian (native)

## SKILLS / AREAS

---

Digital Humans · 3D Modeling and Reconstruction · 3D Scanning · 2D & 3D Computer Vision · 3DMMs · NeRFs · Gaussian Splatting · Neural Rendering · Diffusion Models · Hair and fur reconstruction · Physics-based simulations · Computer Graphics for Vision · Reinforcement Learning · Python · C/C++ · PyTorch · Blender